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CHAPTER 1

Introduction

1.1 Prologue

Automation everywhere in CS, we want to set up boilerplates to help our development.

Examples:

- Continuous integration
- Macros are everywhere
- Artificial intelligence

We delegate tasks to some tools.

expert system →

example plus frappant:

artificial intelligence: based on probability but not reproducible, not predicatable

several other examples in computer sciences:

...

1.2 Automation is everywhere

In the domain of formal proof automation is everywhere, and it is formally called elaboration ??

In formal proof, proofs are verbose and have lot of details: types, coercions, canonical structures, type-classes.

In theorem provers the user benefit of notations, implicit inference and tactics to ease his job.

Simply typed lambda calculus is an example of automation: an algo checks if ...

1.3 Contributions

In this manuscript we focus on an alternative type-class solver written in the elpi meta-language.

We also talk about a formalized tool...

1.3.1 Alternative type-class solver in Coq-Elpi

1.3.2 Static determinacy checking for Elpi

CHAPTER 2

State of the Art

At the very base of my PhD there are two softwares, the Rocq Proof assistant (formerly known as *Coq* [5]) and the *Elpi* [12].

2.1 The *Rocq* proof assistant

[5]

Theorem provers with HO logic programming languages: [14]

Mathematical components: [31]

Abella: [20]

SSReflect: [21]

Isabelle: [37]

2.1.1 The basis of the language

2.1.2 Elaboration

In fig. 2.1 (gently offered by Enrico), we have a skematal representation of the elaboration process.

2.1.3 Ad-hoc polymorphism: Type Classes

One of the most interesting fragment of elaboration is [50, 46, 52]

Cite Haskell [24], in caml it does not exists: `+. vs +`

IRIS: [28]

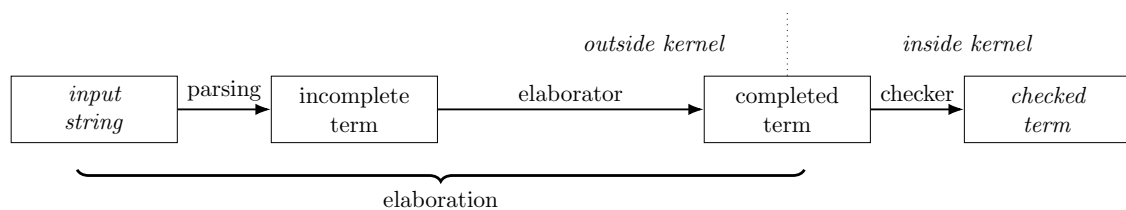


Figure 2.1: Elaboration skeme

2.2 The *Elpi* language

Elpi [12] is a dialect of λ prolog [33, 36]

Teyjus: [35, 34]

Elf: [39]

Twelf: [40]

Lambda prolog: [43]

Mercury: [45]

WAM: [3]

Prolog semantics: [10]

2.2.1 λ -terms representation

HOAS: [38]

2.2.2 Hints about unification

Miller: [33]

Huet: [27]

Modes: [53]

2.2.3 Controlling backtracking: the ! operator

SWI prolog: [47]

Andrews: [2]

Vink: [9]

2.2.4 *CHR*: goal suspension

[19, 18]

CHR elpi: [23, section 4.3]

CHR Pfenning: [32]

2.3 *Coq-Elpi*: Rocq and Elpi interleaving

Coqelpi: [48]

Hierarchy Builder: [8]

Trackt: [6]

Equality test [22]

Derive: [49]

2.3.1 Concluding remarks

Overall, Elpi is good for elaboration

CHAPTER 3

Alternative type-classes solver in *Coq-Elpi*

Typeclasses are an elegant way to implement ad-hoc polymorphism [51], so that, the same symbol can be reused in different context while keeping the same mathematical properties. From an implementation point of view, typeclasses are

3.1 Why typeclass resolution in *Elpi*?

[15]

Rule base automation!

3.2 From Rocq goals to *Elpi* proofs

3.2.1 Rule encoding

The hooks

3.2.2 Goal encoding

3.3 Alternative search indexing

3.3.1 Discrimination trees

Discrimination trees: Pfenning notes *

TODO: do some benchmarks

Discrimination trees: opt for lists and tree depths.

3.3.2 Some ideas on tabling

[44, 41]

*<https://www.cs.cmu.edu/~fp/courses/99-atp/lectures/lecture28.html>

3.4 Helping higher-order unification

[16]

?: [13]

Variadic functions are typechecked in coq [7]

3.4.1 η -equivalent terms

3.4.2 β -equivalent terms

[1]

Beta problema outside the pattern can come back in LL [35]

3.5 Experience report

IRIS: [28]

CHAPTER 4

Determinacy checking for Elpi

4.1 Why determinacy?

[11, 45, 25, 26, 30, 29]

4.2 Implementation

4.2.1 Rule checking

4.2.2 Mutual exclusion

Using discrimination tree

4.2.3 Caveats

For example, mutual exclusion is not perfect: over approximation of unification

Rule checking ignores flexible rule loading

4.2.4 Experience report

4.3 Formalization in Rocq

Using ssreflect [21]

4.3.1 The language

Input must come before outputs

Inputs must come before outputs, with Disc tree this is not mandatory, since unification is weaker.

EG:

```
pred g o:int, o:int.  
  
f 3 3 (g X X).  
f 3 4 (g 3 4).  
  
main:-  
  f X Y (g X Y).
```

In the code above, both rules can be used with the query, but, the discrimination tree, save the world

4.3.2 The tree semantics

4.3.3 The stack semantics

[\[42\]](#)

4.3.4 Semantics equivalence

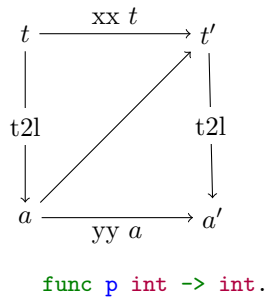
[\[4\]](#)

4.3.5 The main theorem

CHAPTER 5

Conclusion and Perspectives

[17]



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